

Problem 4, Cell 3: Sun's conjecture for all $k \leq 8$

Official statement. For integers a and $m \geq 1$ write $a \pmod{m} = \{a + mt : t \in \mathbb{Z}\}$. A finite family of classes $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ is pairwise disjoint if $a_i \pmod{m_i} \cap a_j \pmod{m_j} = \emptyset$ whenever $i < j$. Nothing is assumed about the moduli beyond $m_i \geq 1$: they may repeat, and the residues a_i may repeat as well. By the Chinese remainder theorem,

$$a_i \pmod{m_i} \cap a_j \pmod{m_j} \neq \emptyset \iff \gcd(m_i, m_j) \mid a_i - a_j. \quad (*)$$

Question. Let $k \geq 2$ and let $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ be pairwise disjoint. Must there be a pair $i < j$ with $\gcd(m_i, m_j) \geq k$?

A first range of sizes: after C1 and C2, the sizes $k = 5, 6, 7$ and 8 remain. Prove the statement for every $k \leq 8$.

Theorem 1 below is exactly this statement, for all $2 \leq k \leq 8$ (the question assumes $k \geq 2$). The notation $a \pmod{m}$ below is the class $a \pmod{m}$ of the statement. Lemma 2 is $(*)$, proved for completeness.

Known result. The question is Sun's disjoint congruence classes conjecture [2]. O'Bryant [1] proved it for all $k \leq 20$, so this case is known. The proof below is an independent proof and is self-contained.

For an integer $m \geq 1$ and $a \in \mathbb{Z}$ write $a \pmod{m} = \{x \in \mathbb{Z} : x \equiv a \pmod{m}\}$. A family of residue classes is *disjoint* if its members are pairwise disjoint.

Theorem 1. *Let $2 \leq k \leq 8$ and let $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ be a disjoint family of residue classes. Then $\gcd(m_i, m_j) \geq k$ for some $i \neq j$.*

The proof is by hand throughout; no computation is used. The cases $k = 5, 6$ follow from Lemma 5, and the cases $k = 7, 8$ from Proposition 9. The same arguments also give $k = 2, 3, 4$ again.

Notation. Throughout, $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ ($k \geq 2$) is a disjoint family, $[k] = \{1, \dots, k\}$, and $g_{ij} = \gcd(m_i, m_j)$ for $i \neq j$. For an integer $n \geq 1$, $\text{pr}(n)$ denotes the set of primes dividing n .

1 Preliminaries

Lemma 2. *Let $m, n \geq 1$, $a, b \in \mathbb{Z}$ and $g = \gcd(m, n)$. Then $(a \pmod{m}) \cap (b \pmod{n}) \neq \emptyset$ if and only if $g \mid a - b$.*

Proof. $(\Rightarrow)$ If $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$, then $a \equiv x \equiv b \pmod{g}$. $(\Leftarrow)$ Choose $u, v \in \mathbb{Z}$ with $um + vn = g$ (Bézout). Write $a - b = gt$ and put $x = a - umt$. Then $x \equiv a \pmod{m}$, and $x = a - (g - vn)t = b + vnt \equiv b \pmod{n}$. $\square$

Corollary 3. *For all $i \neq j$ we have $g_{ij} \geq 2$ and $a_i \not\equiv a_j \pmod{g_{ij}}$. Consequently, if $g_{ij} \mid q$ for some integer q , then $a_i \not\equiv a_j \pmod{q}$.*

Proof. By Lemma 2, disjointness means $g_{ij} \nmid a_i - a_j$, and this forces $g_{ij} \neq 1$. If $a_i \equiv a_j \pmod{q}$ and $g_{ij} \mid q$, then $a_i \equiv a_j \pmod{g_{ij}}$, a contradiction. $\square$

We use the pigeonhole principle in the following form. If integers a_i ($i \in I$) are pairwise incongruent modulo q , then $|I| \leq q$.

Prime channels. For $i \in [k]$ let

$$\pi(i) = \{p \text{ prime} : p \mid g_{ij} \text{ for some } j \neq i\}, \quad S_p = \{i \in [k] : p \in \pi(i)\}.$$

Lemma 4 (Channel lemma). (a) $\text{pr}(g_{ij}) = \pi(i) \cap \pi(j)$ for all $i \neq j$; in particular $\pi(i) \cap \pi(j) \neq \emptyset$.

(b) $|S_p| \neq 1$ for every prime p .

(c) If p is a prime and $i \neq j$ are both in S_p , then $p \mid g_{ij}$.

Proof. (a) The inclusion $\subseteq$ holds by definition of π . Conversely, if $p \in \pi(i) \cap \pi(j)$, then p divides some g_{il} and hence m_i , and similarly $p \mid m_j$; so $p \mid g_{ij}$. The set $\text{pr}(g_{ij})$ is nonempty because $g_{ij} \geq 2$ (Corollary 3). (b) If $i \in S_p$, then $p \mid g_{ij}$ for some $j \neq i$, and then $j \in S_p$ as well. (c) This is (a). $\square$

2 All gcds prime powers: $k \leq 6$

Lemma 5 (Prime-power lemma). Let $2 \leq M \leq 5$ and suppose every g_{ij} is a prime power with $g_{ij} \leq M$. For each prime $p \leq M$ let q_p be the largest power of p with $q_p \leq M$, and let P be the set of primes $\leq M$. Then

$$k \leq \max\left(\max_{p \in P} q_p, \binom{|P|}{2}\right).$$

Proof. By Lemma 4(a), each $\pi(i) \cap \pi(j) = \text{pr}(g_{ij})$ is a single prime, which lies in P . Every prime in $\pi(i)$ divides some $g_{ij} \leq M$, so $\pi(i) \subseteq P$.

Case I: some prime p lies in every $\pi(i)$. Then $p \in \pi(i) \cap \pi(j) = \text{pr}(g_{ij})$ for all $i \neq j$, so every g_{ij} is a power of p that is at most M . Hence $g_{ij} \mid q_p$. By Corollary 3, the a_i are pairwise incongruent modulo q_p , so $k \leq q_p$ by pigeonhole.

Case II: no prime lies in every $\pi(i)$. If some $\pi(i) = \{p\}$, then every $\pi(j)$ ($j \neq i$) meets $\{p\}$, so p lies in all $\pi(\cdot)$, which is Case I. Hence $|\pi(i)| \geq 2$ for all i . Two distinct indices $i \neq j$ cannot satisfy $\pi(i) = \pi(j)$, since then $|\pi(i) \cap \pi(j)| \geq 2$. Also $|\pi(i)| \leq 2$ for every i . Indeed, $|P| \leq 3$ because $M \leq 5$. So $|\pi(i)| \geq 3$ would force $\pi(i) = P \supseteq \pi(j)$ for any $j \neq i$ (such a j exists since $k \geq 2$), and then $|\pi(i) \cap \pi(j)| = |\pi(j)| \geq 2$. Hence the $\pi(i)$ are pairwise distinct 2-element subsets of P , so $k \leq \binom{|P|}{2}$. $\square$

Corollary 6. Theorem 1 holds for $2 \leq k \leq 6$.

Proof. For $k = 2$ this is Corollary 3. Let $3 \leq k \leq 6$ and suppose all $g_{ij} \leq k - 1$. Put $M = k - 1 \in \{2, 3, 4, 5\}$. Every integer in $\{2, \dots, 5\}$ is a prime power, so Lemma 5 applies:

k	M	P	$\max_p q_p$	$\binom{ P }{2}$
3	2	$\{2\}$	2	0
4	3	$\{2, 3\}$	3	1
5	4	$\{2, 3\}$	4	1
6	5	$\{2, 3, 5\}$	5	3

In every row the bound of Lemma 5 equals $k - 1 < k$, a contradiction. $\square$

Remark 7. For $k = 5$ the sets S_2 and S_3 play the role of the sets E and T of Cell 2 (in general only $S_2 \subseteq E$ and $S_3 \subseteq T$). Case II would require two indices with $\pi(i) = \pi(j) = \{2, 3\}$, i.e. $6 \mid g_{ij}$. In Case I with $p = 2$ all gcds divide 4, and with $p = 3$ all gcds equal 3.

3 Gcds in $\{2, 3, 4, 6\}$

Lemma 8. *If every g_{ij} lies in $\{2, 3, 4, 6\}$, then $k \leq 6$.*

Proof. Let $E = \{i : 2 \mid m_i\}$ and $T = \{i : 3 \mid m_i\}$. Each i lies in $E \cup T$, because g_{ij} (for any $j \neq i$) is divisible by 2 or 3 and divides m_i . If $i \in E \setminus T$ and $j \in T \setminus E$, then g_{ij} is coprime to 6, which is impossible. Hence $T \setminus E = \emptyset$ or $E \setminus T = \emptyset$, that is, $[k] = E$ or $[k] = T$. Put $A = E \cap T = \{i : 6 \mid m_i\}$. For $i \neq j$ in A we have $6 \mid g_{ij}$, so $g_{ij} = 6$.

Case $[k] = T$. Let $B = T \setminus E$. If $i \neq j$ and at least one of them is in B , then $3 \mid g_{ij}$ and $2 \nmid g_{ij}$, so $g_{ij} = 3$. By Corollary 3:

- the a_b ($b \in B$) are pairwise incongruent mod 3;
- each a_i ($i \in A$) is incongruent mod 3 to every a_b ($b \in B$);
- the a_i ($i \in A$) are pairwise incongruent mod 6.

So the residues $a_i \bmod 6$ ($i \in A$) are distinct, and they lie in the preimage of the $3 - |B|$ classes of $\mathbb{Z}/3$ not used by B . Each class of $\mathbb{Z}/3$ has exactly two preimages in $\mathbb{Z}/6$, so $|A| \leq 2(3 - |B|)$. Then $k = |A| + |B| \leq 6 - |B| \leq 6$.

Case $[k] = E$. Let $B = E \setminus T$, $A_2 = \{i \in A : 4 \mid m_i\}$ and $A_1 = A \setminus A_2$. We have $|A_2| \leq 1$, since two indices of A_2 would give $12 \mid g_{ij}$. For $b \neq b'$ in B , $g_{bb'}$ is even and not divisible by 3, so $g_{bb'} \in \{2, 4\}$ and $g_{bb'} \mid 4$. For $i \in A_1$ and $b \in B$, g_{ib} is even, not divisible by 3 (as $3 \nmid m_b$) and not divisible by 4 (as $4 \nmid m_i$), so $g_{ib} = 2$. By Corollary 3:

- the a_b ($b \in B$) are pairwise incongruent mod 4;
- each a_i ($i \in A_1$) has parity opposite to every a_b ($b \in B$);
- the a_i ($i \in A$) are pairwise incongruent mod 6.

If $B = \emptyset$, then $[k] = A$ and $k \leq 6$ by pigeonhole mod 6. If the a_b ($b \in B$) have both parities, then $A_1 = \emptyset$ and $k \leq |B| + |A_2| \leq 4 + 1 = 5$. Otherwise all a_b have the same parity ε . Then $|B| \leq 2$, since there are only two classes mod 4 of parity ε . Every a_i with $i \in A_1$ has parity $1 - \varepsilon$, and these a_i are pairwise incongruent mod 6. Only three classes mod 6 have a given parity, so $|A_1| \leq 3$. Altogether $k \leq 2 + 3 + 1 = 6$. $\square$

4 Gcds at most 7: the cases $k = 7, 8$

Proposition 9. *Suppose every $g_{ij} \leq 7$. Then $k \leq 7$. If moreover no g_{ij} equals 7, then $k \leq 6$.*

Proof. For $n \in \{2, \dots, 7\}$ the sets $\text{pr}(n)$ are $\{2\}, \{3\}, \{2\}, \{5\}, \{2, 3\}, \{7\}$. By definition of $\pi(i)$, since every $g_{ij} \leq 7$, every $\pi(i) \subseteq \{2, 3, 5, 7\}$. Let $p \in \{5, 7\}$. For $i \neq j$ in S_p , Lemma 4(c) gives $p \mid g_{ij} \leq 7$, so

$$g_{ij} = p \quad \text{and} \quad \pi(i) \cap \pi(j) = \{p\} \quad (i \neq j \in S_p). \quad (1)$$

In particular the sets $\pi(i) \setminus \{p\}$, $i \in S_p$, are pairwise disjoint.

Step 1. For $p \in \{5, 7\}$: either $S_p = [k]$, in which case $k \leq p$ and every $g_{ij} = p$; or $|S_p| \leq |\pi(l)|$ for every $l \notin S_p$. If $S_p = [k]$, then (1) gives $g_{ij} = p$ for all pairs, and $k \leq p$ by Corollary 3 and pigeonhole. Otherwise, let $l \notin S_p$. For each $i \in S_p$, the set $\pi(i) \cap \pi(l)$ is nonempty (Lemma 4(a)) and contained in $\pi(i) \setminus \{p\}$ (since $p \notin \pi(l)$). These nonempty sets are pairwise disjoint subsets of $\pi(l)$, so $|S_p| \leq |\pi(l)|$.

Let $O = [k] \setminus (S_5 \cup S_7)$. Thus $\pi(l) \subseteq \{2, 3\}$ for $l \in O$.

Step 2. The case $O = \emptyset$. Then $[k] = S_5 \cup S_7$.

- If $S_5 = [k]$, then all $g_{ij} = 5$ and $k \leq 5$.
- If $S_7 = [k]$, then all $g_{ij} = 7$ and $k \leq 7$.
- Otherwise, for each $p \in \{5, 7\}$ some $l \notin S_p$ exists. Since $\pi(l) \subseteq \{2, 3, 5, 7\} \setminus \{p\}$, Step 1 gives $|S_p| \leq 3$. Hence $k \leq |S_5| + |S_7| \leq 6$.

Step 3. The case $O \neq \emptyset$. Fix $l_0 \in O$ and let $p \in \{5, 7\}$ with $S_p \neq \emptyset$. By Step 1 (note $l_0 \notin S_p$), $|S_p| \leq |\pi(l_0)| \leq 2$, and by Lemma 4(b), $|S_p| = 2$. Write $S_p = \{i, i'\}$. For every $l \in O$, the sets $\pi(i) \cap \pi(l)$ and $\pi(i') \cap \pi(l)$ are nonempty, disjoint (by (1)) and contained in $\pi(l) \subseteq \{2, 3\}$. Therefore $\pi(l) = \{2, 3\}$. After renaming i, i' we may take $\pi(i) \cap \pi(l) = \{2\}$ and $\pi(i') \cap \pi(l) = \{3\}$. The names i, i' do not depend on l : we have $\pi(i) \cap \{2, 3\} = \pi(i) \cap \pi(l)$ for every $l \in O$.

Step 3(a): $S_5 = S_7 = \emptyset$. Then $O = [k]$, and every $\text{pr}(g_{ij}) = \pi(i) \cap \pi(j) \subseteq \{2, 3\}$. With $g_{ij} \leq 7$ this means $g_{ij} \in \{2, 3, 4, 6\}$. Lemma 8 gives $k \leq 6$.

Step 3(b): some $S_p = \{i, i'\}$ is nonempty, with notation as above. We derive constraints on the a_l ($l \in O$).

- For $l \neq l'$ in O : $\text{pr}(g_{ll'}) = \{2, 3\}$ and $g_{ll'} \leq 7$, so $g_{ll'} = 6$. The a_l are pairwise incongruent mod 6.
- For $l \in O$: $\text{pr}(g_{i'l}) = \{3\}$, so $g_{i'l} = 3$ and $a_l \not\equiv a_{i'} \pmod{3}$.
- For $l \in O$: $\text{pr}(g_{il}) = \{2\}$, so $g_{il} \in \{2, 4\}$. At most one $l \in O$ has $4 \mid m_l$, since two such l, l' would give $12 \mid g_{ll'}$. For every other $l \in O$ we get $g_{il} = 2$ (as $g_{il} \mid m_l$), hence $a_l \not\equiv a_i \pmod{2}$.

Let $O' = \{l \in O : 4 \nmid m_l\}$, so $|O| \leq |O'| + 1$. The residues $a_l \pmod{6}$ ($l \in O'$) are distinct. Each has parity different from a_i and residue mod 3 different from $a_{i'}$. By the Chinese remainder theorem, $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$, and exactly $1 \cdot 2 = 2$ classes mod 6 meet both conditions. So $|O'| \leq 2$ and $|O| \leq 3$. Step 3 also showed $|S_q| \in \{0, 2\}$ for both $q \in \{5, 7\}$. Hence

$$k \leq |S_5| + |S_7| + |O| \leq 2 + 2 + 3 = 7.$$

Conclusion. In every case $k \leq 7$. Now suppose no g_{ij} equals 7. Then no prime channel is 7, i.e. $S_7 = \emptyset$ by (1) and Lemma 4(b). The cases give:

- Step 2: $k \leq 5$ ($S_5 = [k]$ is forced, because $S_7 = \emptyset$ and $O = \emptyset$);
- Step 3(a): $k \leq 6$;
- Step 3(b): $k \leq |S_5| + |O| \leq 2 + 3 = 5$.

So $k \leq 6$. □

5 Proof of Theorem 1

The cases $2 \leq k \leq 6$ are Corollary 6. Suppose $k = 7$ and all $g_{ij} \leq 6$. Then Proposition 9 (second part) gives $k \leq 6$, a contradiction. Suppose $k = 8$ and all $g_{ij} \leq 7$. Then Proposition 9 (first part) gives $k \leq 7$, a contradiction. $\square$

Remark 10 (Sharpness). For every k the family $r \bmod k$ ($0 \leq r < k$) is disjoint with all pairwise gcds equal to k . The bounds in Lemma 8 and Proposition 9 are attained. The six classes $r \bmod 6$ have all gcds equal to 6. The seven classes $r \bmod 7$ have all gcds equal to 7.

Remark 11 (Computer sanity check, not part of the proof). The script `check_p4_c3.py` checks the theorem numerically; the proof does not use it.

- *Graph.* For each $k \in \{5, 6, 7, 8\}$ it forms all classes $a \bmod m$ with $0 \leq a < m \leq 60$ (1829 classes). Two classes are joined when they are disjoint and their moduli have $\gcd < k$.
- *Disjointness.* This is checked literally, from the definition, and not via $(*)$ or Lemma 2. The classes $a \bmod m$ and $b \bmod n$ meet iff $b - a \equiv mt \pmod{n}$ for some t , and the set $\{mt \bmod n : 0 \leq t < n\}$ is listed explicitly.
- *Result.* An exact branch-and-bound search finds a maximum clique size of exactly $k - 1$ in each case (5.5s in total). This agrees with the theorem.

References

- [1] K. O'Bryant, *On Z.-W. Sun's disjoint congruence classes conjecture*, in *Combinatorial Number Theory*, de Gruyter, Berlin, 2007, pp. 403–411; arXiv:math/0604347.
- [2] Z.-W. Sun, *Finite covers of groups by cosets or subgroups*, *Internat. J. Math.* **17** (2006), 1047–1064; arXiv:math/0501451.